

Inverses for linear functions

$Fx = b$ in coords $Fx = b$

linear func $\swarrow \searrow$ vectors matrix $\swarrow \searrow$ columns

Gaussian elimination steps do not change invertibility
all of these row operations are multiplication on the left of certain matrices

if we can find $\underbrace{B_k \dots B_2 B_1}_{A^{-1}} A = I$

from last class: check $A^{-1} A = I$

$$\begin{bmatrix} 1 & -1 & 0 \\ 0 & 1 & 0 \\ 0 & 0 & 1/7 \end{bmatrix} \begin{bmatrix} 1 & 1 & 6 \\ 0 & 1 & 0 \\ 0 & 0 & 7 \end{bmatrix} = I$$

check $A \circ A^{-1} = I$

$$\begin{bmatrix} 1 & 1 & 6 \\ 0 & 1 & 0 \\ 0 & 0 & 7 \end{bmatrix} \begin{bmatrix} 1 & -1 & 0 \\ 0 & 1 & 0 \\ 0 & 0 & 1/7 \end{bmatrix} = I$$

$$= \begin{bmatrix} 1 & 0 & 0 \\ 0 & 1 & 0 \\ 0 & 0 & 1 \end{bmatrix} \text{ yes so } \begin{bmatrix} 1 & -1 & 0 \\ 0 & 1 & 0 \\ 0 & 0 & 1/7 \end{bmatrix} \text{ is the left and right inverse}$$

Thrm: A has a left inverse iff it has a right inverse
if either is true these inverses are the same

See previous class notes for faster inverse computation

Lemma: if we do row operations to the augmented matrix $[A|I]$ and obtain $[I|B]$ then $B = A^{-1}$

Ex: compute A^{-1} $A = \begin{bmatrix} 1 & 2 \\ 3 & 4 \end{bmatrix}$

$$\begin{array}{c|cc} 1 & 2 & 1 & 0 \\ 3 & 4 & 0 & 1 \end{array} \xrightarrow{\frac{1}{3}r_2} \begin{array}{c|cc} 1 & 2 & 1 & 0 \\ 1 & 4/3 & 0 & 1/3 \end{array} \xrightarrow{r_2 - r_1} \begin{array}{c|cc} 1 & 2 & 1 & 0 \\ 0 & -2/3 & -1 & 2/3 \end{array} \xrightarrow{-\frac{3}{2}r_2} \begin{array}{c|cc} 1 & 2 & 1 & 0 \\ 0 & 1 & 3/2 & -1/2 \end{array}$$

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 $r_1 - 2r_2$

check

$$\begin{array}{c|cc} 1 & 0 & -2 & 1 \\ 0 & 1 & 3/2 & -1/2 \end{array}$$

$$\begin{bmatrix} 1 & 2 \\ 3 & 4 \end{bmatrix} \begin{bmatrix} -2 & 1 \\ 3/2 & -1/2 \end{bmatrix} = \begin{bmatrix} -2+3 & 1-1 \\ -6+6 & 3-2 \end{bmatrix} = \begin{bmatrix} 1 & 0 \\ 0 & 1 \end{bmatrix}$$